Exploration in Reinforcement Learning

From Bandits to Deep Exploration

RL for Real

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3. Multi-Armed Bandits — Online Setting: Optimism & UCB

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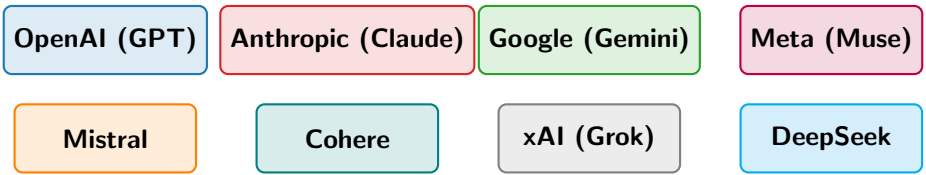
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Part I

Foundations

Choosing an AI Agent

You need the best AI model for your task. There are many options:

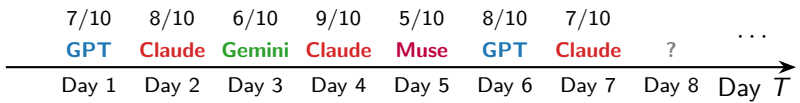


How do you decide which is best?

The Experiment

Be methodical: each day, pick one model, use it, score its quality.

- Each model has an unknown average quality μ_i (plus noise)
- You have a limited budget of T days

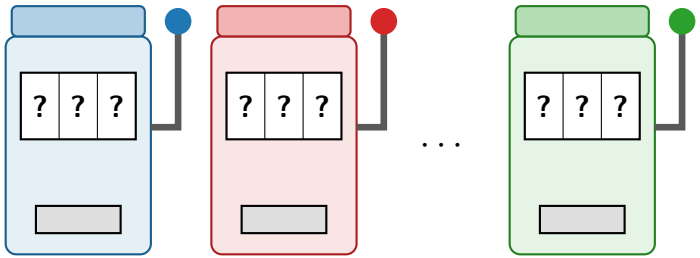


Goal: get the highest total quality over the T days.

This is a variation of the canonical **multi-armed bandit** problem.

The Multi-Armed Bandit Problem

A slot machine is a *one-armed bandit*. Many machines: a **multi-armed bandit**.



Model = arm, day = pull, quality score = reward.

Robbins (1952); see Lattimore & Szepesvári (2020) for a modern treatment.

How Should You Choose?

Random Sampling

Try models uniformly at random

Wastes many days on bad models

Greedy

Always pick the current best

Might miss a better model you tried only once

ε-Greedy

Usually pick the best, sometimes try random

Better, but still undirected

None of these uses uncertainty to direct exploration.

Can we do better?

Before tackling many arms —

let's simplify. What if there are only two?

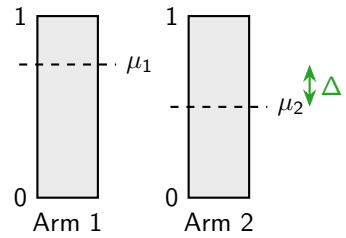
Start offline: assume we have n samples from each arm.

Setup: Two-Armed Bandit

Simplest identification problem:

- Two arms with means μ_1, μ_2
- Gap: $\Delta = |\mu_1 - \mu_2|$
- Rewards bounded in $[0, 1]$
- We have n samples per arm

Goal: identify the best arm from the data.



When Can We Identify the Best Arm?

By Hoeffding's inequality and a union bound, with probability $\geq 1 - \delta$, for both arms:

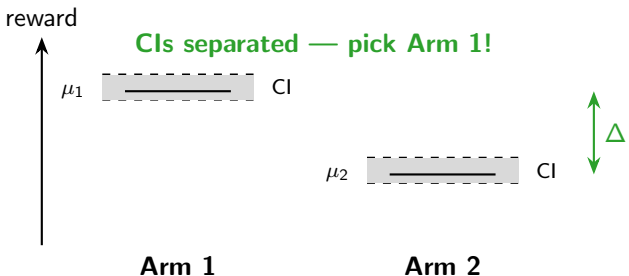
$$\hat{\mu}_i \in \left[\mu_i - \sqrt{\frac{\log(4/\delta)}{2n}},\ \mu_i + \sqrt{\frac{\log(4/\delta)}{2n}} \right]$$

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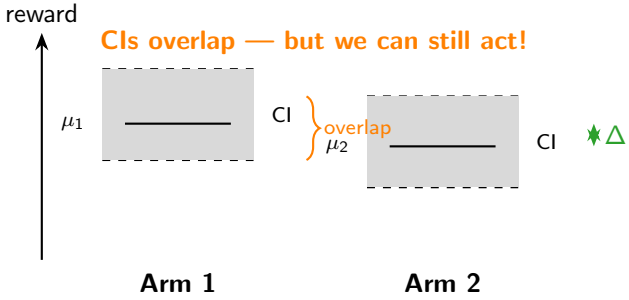
Case 1: Enough samples — CIs don't overlap $\Rightarrow$ we can tell the arms apart!



The empirical best arm $\hat{a} = \text{argmax}_i\, \hat{\mu}_i$ *is* the optimal arm — zero suboptimality!

When Identification Fails

Not enough samples: CIs overlap.



Pick $\hat{a} = \text{argmax}_i \hat{\mu}_i$. The suboptimality is bounded by the CI width:

$$\mu^* - \mu_{\hat{a}} \;\leq\; 2\sqrt{\frac{\log(4/\delta)}{2n}} \;=\; \mathcal{O}\left(\frac{1}{\sqrt{n}}\right)$$

Offline Identification: Summary

Pick the empirical best arm: $\hat{a} = \text{argmax}_i \, \hat{\mu}_i$. How well does it perform?

$$\mu^* - \mu_{\hat{a}} = \begin{cases} 0 & \text{if } n > \dfrac{2\log(4/\delta)}{\Delta^2} \quad \text{(CIs separated)} \\ \mathcal{O}\left(\dfrac{1}{\sqrt{n}}\right) & \text{otherwise} \quad \text{(CIs overlap)} \end{cases}$$

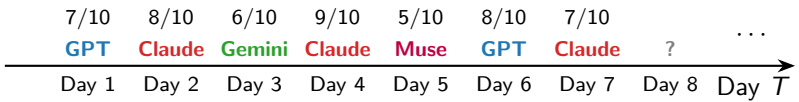
Next question: what if we get to *choose* which arm to sample?

Online Setting

So far the data was given to us. Now *we choose* which model to try each day.

Protocol: at each round $t = 1, 2, \ldots, T$:

1. Pick a model (arm) $a_t \in \{1, \ldots, K\}$
2. Use it and observe a score $r_t \sim \nu_{a_t}$
3. Update your estimates and strategy



Regret: Measuring Online Performance

Performance measure — Regret:

$$R_T = \sum_{t=1}^{T} \mu^* - \mu_{a_t} \quad \text{where } \mu^* = \max_i \mu_i$$

The exploration–exploitation tradeoff:

Exploit

Pick the model that *looks best* so far ⇒ maximize today's score, but risk missing a better option.

Explore

Try a less-tested model to *learn more* ⇒ sacrifice today's score for better future decisions.

Low regret requires balancing both.

Start Simple: Deterministic Arms

We start from a simple setting: **deterministic arms**.

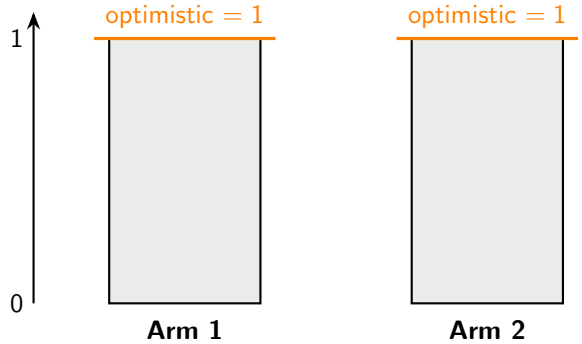
Every pull of arm i returns exactly μ_i — no noise.

Optimism in the Face of Uncertainty (OFU)

When uncertain, assume the best.

OFU — Round 1: No Data Yet

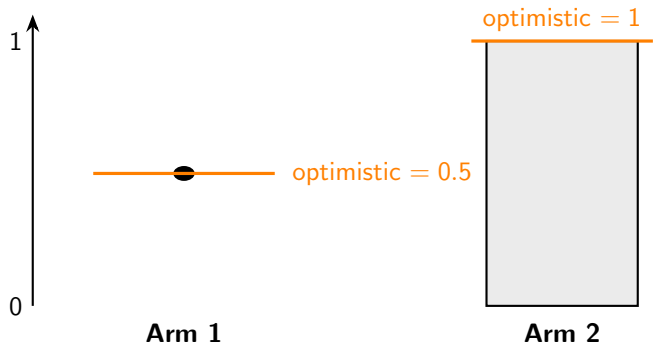
Example: two deterministic arms, $\mu_1 = 0.5$, $\mu_2 = 0.8$ (unknown). Rewards in $[0, 1]$.



Pick either — say Arm 1.

OFU — Round 2: After Observing Arm 1

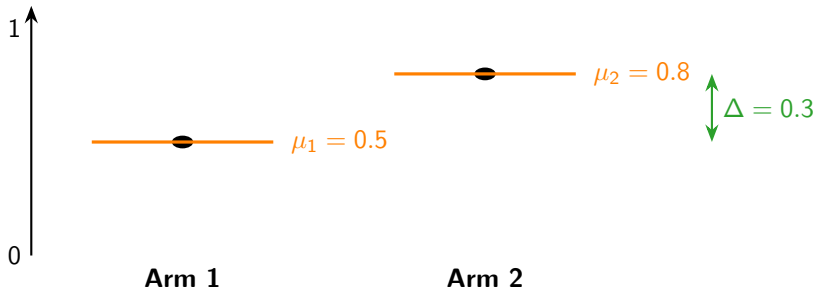
Observed Arm 1: $r = 0.5$ (deterministic $\Rightarrow$ CI collapses to a point).



Optimism $\Rightarrow$ pick Arm 2.

OFU — Both Arms Known

Observed Arm 2: $r = 0.8$. Both CIs collapsed.



After just **2 rounds**, we identified the best arm.

Without optimism, a greedy agent would have stuck with Arm 1 forever.

From Optimism to UCB

What if rewards are *stochastic*?

Be optimistic — but use *confidence intervals*.

UCB Algorithm

UCB1 (Auer et al., 2002): at each round t, select:

$$a_t = \underset{i \in \{1,\ldots,K\}}{\operatorname{argmax}} \underbrace{\hat{\mu}_i(t)}_{\text{estimate}} + \underbrace{\sqrt{\frac{2 \ln t}{n_i(t)}}}_{\text{bonus}}$$

where $\hat{\mu}_i(t)$ = empirical mean, $n_i(t)$ = number of pulls.

Why is this optimistic?

- The bonus is an *upper confidence bound* on the estimation error
- $\hat{\mu}_i(t) + \text{bonus} \geq \mu_i$ with high probability
- We pick the arm with the highest *optimistic* estimate — just like OFU!

UCB: Properties

Exploration bonus shrinks:

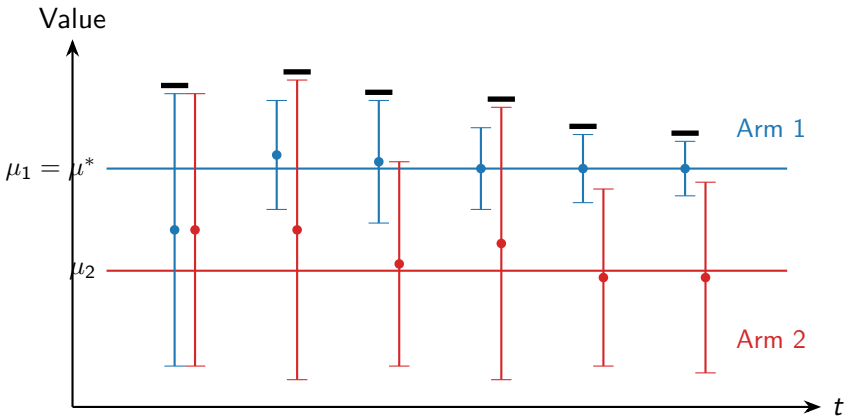
- More pulls ($n_i \uparrow$) $\Rightarrow$ smaller bonus
- Well-explored arms are chosen only if genuinely good

Exploration bonus grows:

- As t increases ($\ln t$), rarely pulled arms get a relative boost
- No arm is ignored forever

Key feature: no tuning parameters — balances exploration and exploitation automatically.

UCB: Confidence Intervals Shrinking Over Time



Bar = highest upper bound $\Rightarrow$ the arm UCB picks. Arm 2's bonus grows while it is ignored, so it is still picked *occasionally* — $\mathcal{O}(\log T)$ times.

UCB: Regret Bound

Theorem (UCB1 Regret)

For K arms with rewards in $[0, 1]$, the UCB1 algorithm achieves:

$$\mathbb{E}[R_T] = \mathcal{O}\Big(\sum_{i:\Delta_i > 0} \frac{\ln T}{\Delta_i} + K \sum_{i=1}^{K} \Delta_i \Big).$$

In the worst case over the gaps, this also gives $\mathbb{E}[R_T] = \mathcal{O}\Big(\sqrt{KT \ln T}\Big)$.

Interpretation:

- Regret is $\mathcal{O}(\log T)$ for fixed gaps, $\mathcal{O}(\sqrt{KT \ln T})$ in the worst case — vs. linear for non-adaptive strategies
- Each suboptimal arm i is pulled only $\mathcal{O}(\ln T / \Delta_i^2)$ times
- Hard-to-distinguish arms (small Δ_i) are pulled more — sensible!

UCB Regret: Proof Sketch 1

Key idea: replace μ^* with the UCB of the chosen arm (optimism!).

Step 1: By optimism, w.h.p. $\mu^* \leq \text{UCB}(a^*)$. Since UCB picks $a_t = \text{argmax}_i \, \text{UCB}(i)$:

$$\mu^* \;\leq\; \text{UCB}(a^*) \;\leq\; \text{UCB}(a_t) \qquad \longleftarrow \text{this is where optimism helps!}$$

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Step 2: Substitute into the regret; w.h.p. $\hat{\mu}_{a_t} \leq \mu_{a_t} + \text{bonus}$, so:

$$R_T = \sum_{t=1}^{T} (\mu^* - \mu_{a_t}) \;\leq\; \sum_{t=1}^{T} \big(\text{UCB}(a_t) - \mu_{a_t}\big) \;\leq\; \sum_{t=1}^{T} 2\sqrt{\frac{2 \ln t}{n_{a_t}(t)}}$$

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Step 3: Cauchy–Schwarz: $\displaystyle\sum_{t=1}^{T} \sqrt{\tfrac{1}{n_{a_t}(t)}} \;\le\; 2\sqrt{KT}$

(Each arm's contribution: $\sum_{j=1}^{n_i} 1/\sqrt{j} \le 2\sqrt{n_i}$. Sum over K arms + Cauchy–Schwarz.)

Result: $R_T = \mathcal{O}\left(\sqrt{KT \ln T}\right)$

UCB Regret: Proof Sketch 2 (1/3)

Key idea: bound how often each suboptimal arm i is pulled.

Step 1: Decompose regret by arm:

$$\mathbb{E}[R_T] = \sum_{i:\Delta_i>0} \Delta_i \cdot \mathbb{E}[n_i(T)]$$

Good event G_t: all confidence intervals are valid at time t,

$$G_t = \left\{ |\hat{\mu}_j(t) - \mu_j| \leq \sqrt{\tfrac{2 \ln t}{n_j(t)}} \text{ for all arms } j \right\}$$

By Hoeffding and a union bound, $\Pr(G_t^c) \leq 2Kt^{-3}$.

UCB Regret: Proof Sketch 2 (2/3)

Key idea: bound how often each suboptimal arm i is pulled.

Step 2: UCB picks arm i at time t only if

$$\text{UCB}(i) \geq \text{UCB}(a^*), \qquad \text{UCB}(j) = \hat{\mu}_j(t) + \sqrt{\tfrac{2 \ln t}{n_j(t)}}$$

Step 3: On G_t, $\text{UCB}(a^*) \geq \mu^*$ and $\hat{\mu}_i \leq \mu_i + \sqrt{2 \ln t / n_i(t)}$. So if arm i is picked:

$$\mu_i + 2\sqrt{\tfrac{2 \ln t}{n_i(t)}} \;\geq\; \text{UCB}(i) \;\geq\; \text{UCB}(a^*) \;\geq\; \mu^*$$

Rearranging with $\Delta_i = \mu^* - \mu_i$:

$$\boxed{n_i(t) \;\leq\; \frac{8 \ln T}{\Delta_i^2}}$$

On the good event, a suboptimal arm stops being pulled once it has enough samples.

UCB Regret: Proof Sketch 2 (3/3)

Key idea: bound how often each suboptimal arm i is pulled.

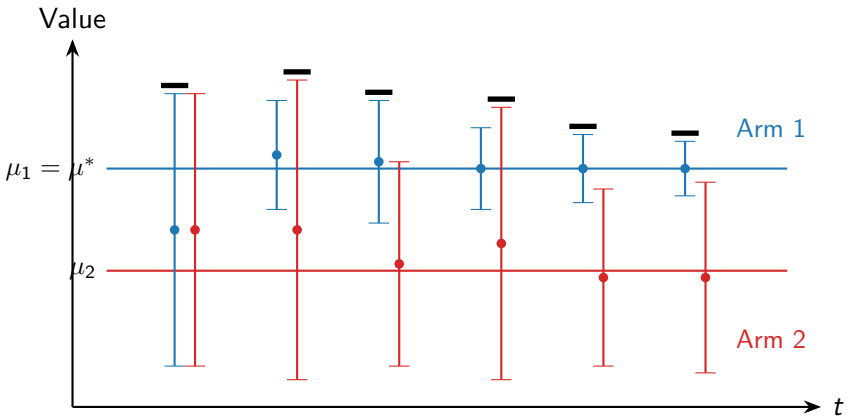
Step 4: Split each round into G_t and G_t^c. Off the good event, charge one pull to arm i:

$$\mathbb{E}[n_i(T)] \;\le\; \frac{8\ln T}{\Delta_i^2} + \sum_{t=1}^{T} \Pr(G_t^c) \;\le\; \frac{8\ln T}{\Delta_i^2} + \sum_{t=1}^{\infty} 2Kt^{-3} \;=\; \frac{8\ln T}{\Delta_i^2} + \mathcal{O}(K)$$

Result: plug into Step 1:

$$\boxed{\mathbb{E}[R_T] \;=\; \mathcal{O}\Big(\sum_{i:\Delta_i>0} \frac{\ln T}{\Delta_i} + K\sum_i \Delta_i\Big)}$$

UCB: Confidence Intervals Shrinking Over Time



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Part II

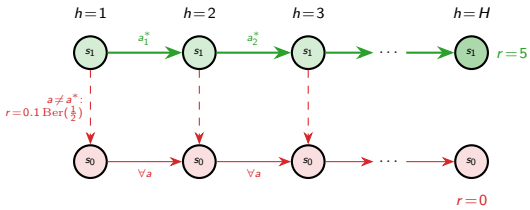
From Theory to Practice

Why Random Exploration Fails

In RL, the agent must find the right sequence of actions.

- Random / ε-greedy succeeds w.p. $|\mathcal{A}|^{-H}$ per episode
- $\Rightarrow \Omega(|\mathcal{A}|^H)$ episodes — **exponential in H**
- We need *directed* exploration

Combination Lock



Wrong action $\Rightarrow$ stuck in s_0 forever.

Optimism in MDPs: Optimistic Q-Learning

Idea (Jin et al., 2018): add a UCB bonus *inside* the Q-learning update, then act greedily.

$$\boxed{Q_h(s,a) \leftarrow (1-\alpha_n)\, Q_h(s,a) + \alpha_n \big[\, r + V_{h+1}(s') + b_n \,\big]} \qquad b_n \propto \tfrac{1}{\sqrt{n}}$$

$n = n_h(s,a)$ is the visit count.

- Rarely visited $(s,a) \Rightarrow$ large bonus
- Bonus propagates through Bellman backups $\Rightarrow$ **deep** exploration
- Regret $\widetilde{\mathcal{O}}\big(\mathrm{poly}(H)\sqrt{SAT}\big)$ — polynomial, not exponential

Large State Spaces Break Counts

The regret bound scales with $\sqrt{S}$. In real problems:

Domain	**State space $\lvert\mathcal{S}\rvert$**
Small grid (10×10)	100
Chess	$\sim 10^{47}$
Atari (pixels)	$\sim 256^{210\times160\times3}$
Robotics (continuous)	∞

- States rarely repeat $\Rightarrow$ counts $n(s, a)$ are almost always 0
- We need a bonus that **generalizes across similar states**

How do we measure novelty without counting?

Exploration Bonuses

Add an *intrinsic reward* that is large on novel states:

$$r_{\text{total}}(s, a) = r_{\text{ext}}(s, a) + \beta\, r_{\text{int}}(s)$$

Method	**Bonus** $r_{\text{int}}(s)$
Counts (UCB)	$1/\sqrt{n(s)}$
Pseudo-counts (Bellemare '16)	$1/\sqrt{\hat{n}(s)}$, from a density model
ICM (Pathak '17)	forward-model prediction error
RND (Burda '19)	$\|\hat{f}_\theta(s) - f^*(s)\|^2$

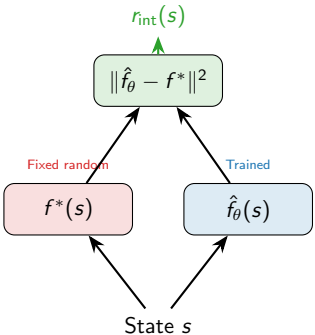
All give a high bonus where the agent has little data $\Rightarrow$ **optimism**.

Random Network Distillation (RND)

Burda et al., 2019

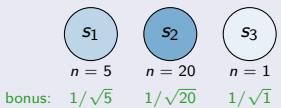
1. **Target** $f^* : \mathcal{S} \to \mathbb{R}^d$ — random network, **fixed**
2. **Predictor** $\hat{f}_\theta : \mathcal{S} \to \mathbb{R}^d$ — trained on visited states to match f^*
3. **Bonus:**

$$\boxed{r_{\text{int}}(s) = \|\hat{f}_\theta(s) - f^*(s)\|^2}$$



RND as Optimism: Prediction Error as a Count

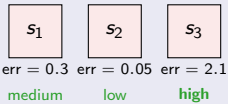
Tabular: Count-Based



$$r_{\text{int}}(s) = 1/\sqrt{n(s)}$$

Requires enumerable states

Neural: Prediction Error



$$r_{\text{int}}(s) = \|\hat{f}_\theta(s) - f^*(s)\|^2$$

Works for any state space

Familiar states $\Rightarrow$ low error. Novel states $\Rightarrow$ high error.

Same optimism as UCB — but it generalizes across states.

Prediction Error $\approx$ Counts: A Linear Model

Model:

- Targets $y = w^{*\top}\phi(s) + \varepsilon$, noise variance σ^2
- Fit $\hat{w}$ by least squares on visited states
- Covariance $\Sigma = \sum_t \phi(s_t)\phi(s_t)^\top$

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Error at s:

$$\mathbb{E}\Big[|(\hat{w} - w^*)^\top \phi(s)|^2\Big] \;=\; \sigma^2 \,\|\phi(s)\|^2_{\Sigma^{-1}} \;\overset{\text{tabular}}{=}\; \frac{\sigma^2}{n(s)}$$

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But one error is a single noisy draw. With d independent targets, the sum concentrates:

$$\sum_{j=1}^{d} |(\hat{w}_j - w_j^*)^\top \phi(s)|^2 \;\approx\; \boxed{\frac{d\,\sigma^2}{n(s)}}$$

RND's $f^* : \mathcal{S} \to \mathbb{R}^d$ provides the d targets.

RND Bonus: Matching UCB

Under this model, $\|\hat{f}(s) - f^*(s)\|^2 \approx d\sigma^2/n(s)$. UCB's bonus is $1/\sqrt{n} \Rightarrow$ use the **norm**:

$$r_{\text{int}}(s) = \|\hat{f}_\theta(s) - f^*(s)\| \propto \frac{1}{\sqrt{n(s)}}$$

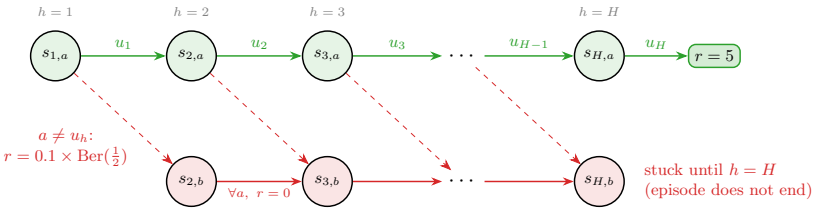
Caveats

- Standard RND (and our notebook) uses $\|\cdot\|^2$
- RND's targets are noiseless; the analysis needs noise $\Rightarrow$ perturb the targets during training

Experimental Setup

ε-greedy DQN vs. RND + DQN on a **diabolical combination lock** (Misra et al., 2020):

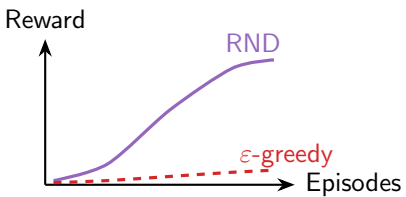
- One correct action per step; a wrong action traps the agent until step H
- Misleading reward for entering the bad chain; noisy observations



Observation (state is never seen): $x = R\left([\, e_s;\ e_h \,] + \varepsilon\right)$, $\varepsilon \sim \mathcal{N}(0,\, 0.1^2 I)$, R: Hadamard matrix (± 1 entries)
Random actions succeed with probability $|\mathcal{A}|^{-H}$.

Results

- ε-greedy: fails for $H \geq 5$
- RND: explores systematically, solves longer horizons

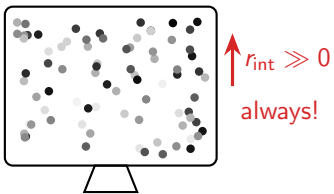


Schematic — see notebook for actual curves.

The Noisy TV Problem

- A TV showing random static: every frame is *novel*
- Prediction error never shrinks $\Rightarrow$ the bonus stays high
- The agent watches TV instead of exploring

Open problem: how to filter out *irrelevant* information when exploring — still not well understood, in practice or in theory (e.g., exogenous noise: Efroni et al., 2022).



Exploration in LLMs: Open Questions

- How can we train LMs to produce outputs that are both **high quality** and **diverse**?
- How can we train agents to **explore** properly in **long-horizon** agentic tasks?
- What does exploration look like in **vision-language models**?

. . . and many more.

Summary

1. **Optimism (tabular):** UCB is near-optimal
   - Not covered: Thompson sampling, information-directed sampling, ...
2. **Deep RL:** uncertainty metrics as a bonus (e.g., RND)

$$\text{value for acting} = \text{estimate} + \beta \cdot \text{uncertainty}$$

3. **Open questions:**
   - Deep RL exploration is ad hoc: can it be principled and general?
   - Exploration in *long-horizon, real-world* tasks (robotics, automated research, ...)

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Thank You!

Questions?

See the companion notebook for full RND implementation and experiments.